

Dayananda Sagar University



SCHOOL OF ENGINEERING

School of Engineering

Department of Computer Science and Engineering

Design and Analysis of Algorithms

Laboratory Manual

Department of Computer Science and Engineering

Design and Analysis of Algorithms

The laboratory lab book belongs to:

Name:	
University Serial Number:	

Vision and Mission of the University

Vision

To be a Centre of excellence in education, research & training, innovation & entrepreneurship and to produce citizens with exceptional leadership qualities to serve national and global needs.

Mission

To achieve our objectives in an environment that enhances creativity, innovation and scholarly pursuits while adhering to our vision.

Values

- The values that drive DSU and support its vision: The Pursuit of Excellence
- A commitment to strive continuously to improve ourselves and our systems with the aim of becoming the best in our field.

Fairness

A commitment to objectivity and impartiality, to earn the trust and respect of society.

Leadership

A commitment to lead responsively and creatively in educational and research processes.

Integrity and Transparency

A commitment to be ethical, sincere and transparent in all activities and to treat all individuals with dignity and respect.

Department of Computer Science & Engineering

VISION

To be recognized as a department of eminence in Computer Science and Engineering focusing on sustainability, inclusive technologies, and societal needs.

MISSION

The Department of Computer Science and Engineering is committed to:

M1: Impart quality technical education by designing and delivering contemporary Computer Science Engineering curricula while emphasizing leadership, ethics, values and integrity.

M2: Transform professionals into technically competent through industry-academia collaboration and innovation ecosystem.

M3: Prepare Computer Science and Engineering graduates to meet ever-growing societal needs.

Program Educational Objectives (PEO's)

After few years of graduation, the graduates of Computer Science and Engineering will be able to:

PEO1: Apply appropriate theory, practices and tools in the design, implementation, maintenance and evaluation of computing in the work place or in higher education.

PEO2: Exhibit professional skills in solving challenging problems in their career and advance to leadership roles.

PEO3: Become effective innovator, researcher, and entrepreneur to provide technical solutions for socio-economic challenges.

Programme Outcome (PO's)

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research- based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSO's)

PSO1: Develop, Analyse, Review and Contribute to efficient, secure and high quality design, implementation, testing and operations of computing system

PSO2: Find and articulate digital and intelligent solution that can fully or partially automate various aspects of human activity.

Dayananda Sagar University



SCHOOL OF ENGINEERING

Laboratory Certificate

This is to certify that Mr./Ms _____ bearing
University Seat number (USN) _____ has satisfactorily completed the
experiments in Design and Analysis of Algorithms Laboratory prescribed by the University for
the 4th semester B.Tech program during the year 2023-2024.

Date: _____

Marks	
Maximum	Obtained

Signature of the Faculty In charge

Signature of the Chairman

Laboratory Environment

- This course will use Ubuntu distribution of Linux as the Operating System for all the exercises
- The tool chain used will be GNU 'gcc' suite of tools
- Editor recommended is 'gedit' which is available in these systems
- 'terminal' program is used to run 'gcc' and the executable code

Basic Instructions to use the tools

1. If the system is not powered on, please turn the system ON by pressing the power button on the PC.
2. Check if the monitor needs to be turned on as well
3. If presented with a boot time option, please select 'Ubuntu' in the list
4. Once Ubuntu has booted up, click on the program selection icon on the bottom left of the screen (a small square with dots).
5. Type 'terminal' in the search bar at the top. The icon for the 'terminal' program will appear below, and click the same to start the terminal program
6. The terminal program will typically be a black coloured window with a prompt to type commands.
7. Type 'gedit <prog_name.c>' on the terminal window.
8. This will open the gedit window, which can be used to type the program. Note that the name of the file you specified (prog_name.c) will appear on the top of the window. If there is a '*' character before the file name, it indicates that the file has not been saved yet. Please remember to save the file once you have finished typing the program.
9. You can close the 'gedit' window to get back to the terminal prompt.
10. Type 'gcc <prog_name.c>' and press enter.
11. If there are no errors/warnings, you will see the command prompt again. Else, 'gcc' will print the errors and warnings with line numbers to help you fix the errors.
12. If there are errors/warnings, fix them and re-run 'gcc' till you get a clean compile.
13. 'gcc' by default will save the executable code in a file by name 'a.out'.
14. To run your program and check the output, please type './a.out' and press enter. This will make the
1. OS run the code and you can check the output.
15. If the code does not give the desired output, or fails with some errors, repeat the edit-compile-test cycle by going back and editing the program using 'gedit'.
For example, if you want to name the file as lab1.c, the following are the commands
gedit lab1.c
gcc lab1.c
./a.out

Department of Computer Science and Engineering

Academic Year 2023-2024/ 4th Semester

Desing and Analysis of Algorithms Laboratory

Index

#	Date	Program list	Total (10M)
1		Linear Search and Binary Search.	
2		Implement the Merge Sort algorithm.	
3		Verification of connected graph using DFS.	
4		Traverse a given graph using BFS	
5		Implementation of Warshalls Algorithm to find Transitive Closure of a given directed graph.	
6		Implement Floyd's algorithm to solve all pairs shortest paths problem	
7		Implement 0/1 Knapsack problem using Dynamic Programming	
8		Implement Kruskal's algorithm to find the minimum cost spanning tree	
9		Implement Dijkstra's algorithm to determine the Single Source Shortest Paths.	
10		Open ended problems	